Mukki Prasanth Raju

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.56
A.P.T.W.R Junior College G.Madugula	Class XII (Board of Intermediate Education Andhra Pradesh)	2019	9.84%
C A High School Munchangiput	Class X (Board of Secondary Education Andhra Pradesh)	2017	7.80%

EXPERIENCE

APGENCO Training Institute

- Completed a mini-project on thermal power plant familiarization as part of an engineering training program.
- Gained on-site exposure to power plant operations across various sections, from coal handling to the electrical division.

PROJECTS

Topology Optimization of a Three-Dimensional Soft Gripper

- Apply nonlinear finite element analysis (FEA) and mathematical modeling to analyze large deformations in soft gripper.
- Develop MATLAB code for topology optimization to determine material distributions for desired shapes and motion trajectories.
- Perform simulations in ANSYS and evaluate the optimized gripper's performance through 3D printing.

Optimization of Facility Layout to Minimize Material Handling Costs

- Implemented the Particle Swarm Optimization (PSO) algorithm to optimize facility layout and minimize material handling costs.

Implementation and Verification of Hyperelastic Material Models in Abaqus

- Developed UMAT and UHYPER subroutines for 3rd-order Ogden and Mooney-Rivlin hyperelastic models, respectively.
- Validated results against Abaqus built-in models by comparing stress and displacement contours under 50% tensile deformation and 90° torsion of a bottom-fixed cube.

Vibration Analysis and Balancing of a Single-Cylinder Engine Chassis System

- Analyzed vibration forces in a 149.5 cc, four-stroke, DTS-i, air-cooled, single-cylinder internal combustion engine.
- Simulated engine forces with and without balance shafts and evaluated the vibration reduction achieved with vibration isolation mounts.

Numerical Estimation of Stress and Refractive Power Maps in Healthy and Keratoconus Eyes

- Numerically estimated stress and refractive power maps for healthy and keratoconus eyes.
- Compared deformed and undeformed eye geometries and analyzed the resulting stress distributions.

Topology Optimization Techniques for Soft Robots

- Reviewed topology optimization methods and algorithms for soft robotic designs across various actuation mechanisms and materials.

- Identified research gaps, key challenges, and future research directions in soft robot design optimization.

SKILLS

Technical Skills: MS Word, MS Excel, MS PowerPoint

Simulation Tools: MATLAB, Abaqus

CAD Tools: SolidWorks, AutoCAD

Soft Skills: Teamwork, Time Management, Adaptability

Languages: English, Telugu, Hindi

RELEVANT COURSES

Finite element method, Topology optimization, Vibrations and dynamics, Advanced mechanics of solids, Machine learning and its applications.

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Vibrations and Dynamics Laboratory: Explained theoretical concepts and demonstrated laboratory experiments to final-year B.Tech. students.

Study Leader: Maintained discipline and supported an effective learning environment during school study hours.

EXTRACURRICULAR

Playing Badminton, Volleyball, Running, Watching sports and Cycling